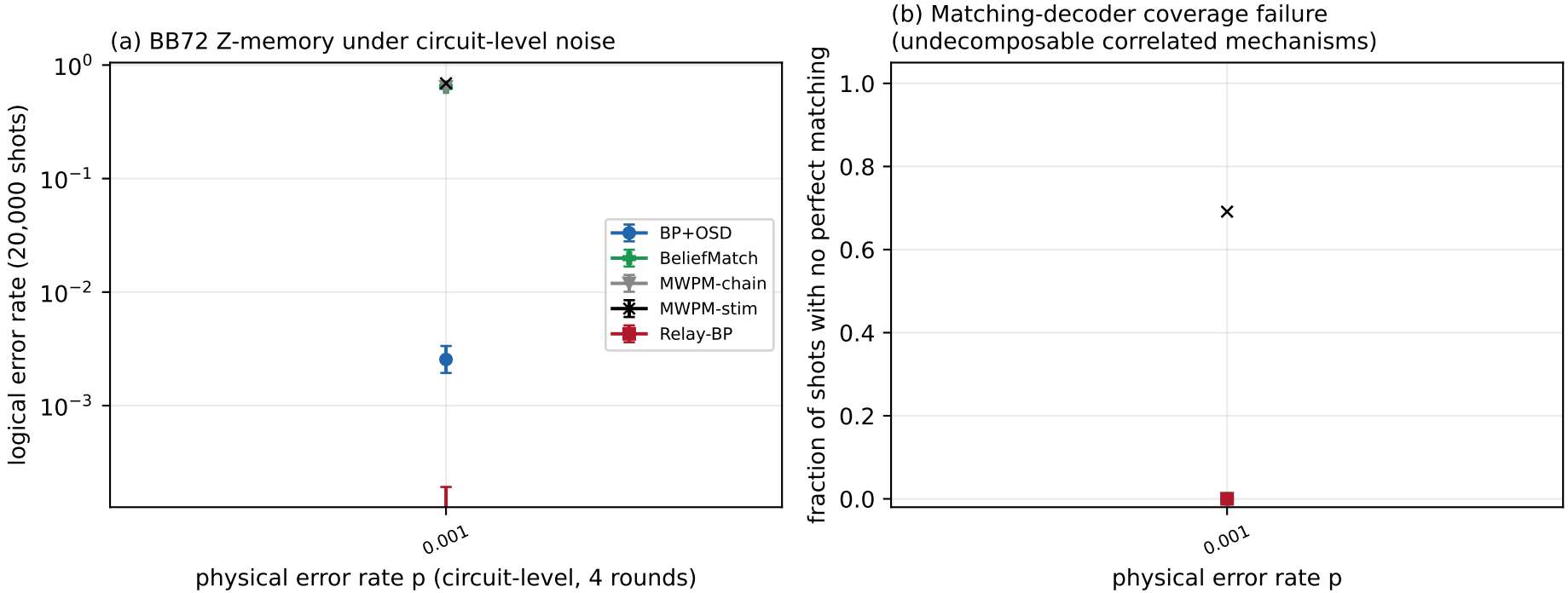
(a) BB72 Z-memory under circuit-level noise
logical error rate (20,000 shots)
10^{0}
10^{-1}
10^{-2}
10^{-3}
BP+OSD
BeliefMatch
MWPM-chain
MWPM-stim
Relay-BP
0.001
physical error rate p (circuit-level, 4 rounds)
(b) Matching-decoder coverage failure
(undecomposable correlated mechanisms)
fraction of shots with no perfect matching
1.0
0.8
0.6
0.4
0.2
0.0
0.001
physical error rate p